Higher National Diploma in Engineering (Civil/Electrical/Mechanical)
Higher National Diploma in Building Service Engineering
IT 1101-Information Technology I
Higher National Diploma in Quantity Surveying
QS 11053-Information Technology I

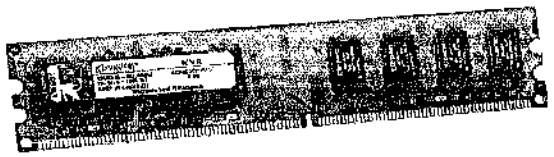
First Year, First Semester Examination – 2018

Instructions for Candidates:	No. of questions	: 06
Answer only **five (05)** questions	No of pages	: 05
All questions carry equal marks	Time	: Three (03) hours

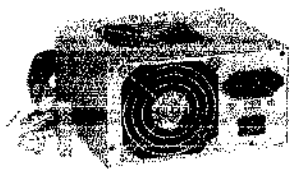
Question 01

i. Identify following devices with the function/purpose of each. **(06 Marks)**

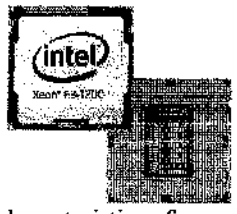
a)



b)



c)



ii. List three (03) characteristics of super computers. **(03 Marks)**

iii. Identify main components of a central processing unit (CPU) and briefly describe them. **(06 Marks)**

iv. Briefly explain what the cache memory is and how it effects to the performance of the computer. **(02 Marks)**

v. Write three (03) factors that you need to consider when you are going to upgrade your current processing unit. **(03 Marks)**

Question 02

i. Describe volatile and non-volatile memory. Give two examples. **(05 Marks)**

ii. Suggest a suitable operating system for following scenarios **(03 Marks)**

a. An operating system for a laptop who doing normal works

b. An operating system for a server computer

c. A smart phone

iii. Explain difference between public domain/open source software and commercial software, give one example for each. **(03 Marks)**

iv. List three (03) usage of MS Word **(03 Marks)**

v. Compare RAM and Hard Disk Drive as storage devises **(06 Marks)**

	RAM	Hard Disk
Capacity		
Speed		
Cost per Mega Byte		
Storage technology		
Volatile/ nonvolatile		
Power consumption		

Question 03

Bellow table displays numbers of computers belongs to each ATIs.

Galle	Colombo	Jaffna
5	10	6

i. Assume that you need to design a computer network for ATI Galle, write down the network topology that you are going to use and justify your answer. **(03 Marks)**

ii. List three (03) networking devises/components or equipment that you need to design above network. **(03 Marks)**

iii. Draw a sketch of your network design with all components. **(04 Marks)**

iv. If you interconnect ATI Galle network with ATI Jaffna network, which data transmission media is/are most suitable. Justify your answer. **(03 Marks)**

v. Briefly describe the function of a device that can be used to provide the security of the network. **(04 Marks)**

vi. Using a diagram show how to connect that above mentioned device to the network **(03 Marks)**

Question 04

i. In an excel sheet, formula of cell C2 is =A$1+B2. When you copy that formula to C3 write the formula in C3. **(02 Marks)**

A part of a payroll of Udugama Maha Vidyala is listed bellow

	A	B	C	D	E	F	G	H	I
1									
2	EPF	12%							
3	TAX	30%							
4									
5			Earnings				Deductions		
6	Name	Designation	Basic Salary	C.O.L	Special Allowance	Total Earning	EPF	TAX	NET PAY
7	Sunil	Teacher	25,000.00	7,800.00	7,500.00	40,300.00	4,836.00	150.00	35,314.00
8	Kamal	Clerk	20,000.00	7,800.00	0.00	27,800.00	3,336.00	0.00	24,464.00
9	Amal	Pion	15,000.00	7,800.00	0.00	22,800.00	2,736.00	0.00	20,064.00
10	Raja	Clerk	20,000.00	7,800.00	0.00	27,800.00	3,336.00	0.00	24,464.00
11	Renu	Teacher	25,000.00	7,800.00	7,500.00	40,300.00	4,836.00	150.00	35,314.00
12									

	Number of employees	
13		
14	Clerk	2
15	Teacher	2
16	Pion	1

Using ***cell references only***, write formula/functions for the followings.

i. To display special allowance for the cell E7. (all teachers are eligible for Rs.7500.00 of allowance) **(03 Marks)**

ii. To display total earning in the cell F7. **(03 Marks)**

iii. 15% (in the cell B2) of total earning will be deducted as EPF. Write a formula for the cell G7. **(03 Marks)**

iv. The first Rs. 40000.00 from their total earning is tax free. For the rest they pay 50% (in the cell B3) as tax. Write a formula for the cell H7 to display tax. **(03 Marks)**

v. To display net pay in the cell I7. **(03 Marks)**

vi. To display numbers of clerk in the cell B14. **(03 Marks)**

Question 05

i. List two (02) examples for DBMSs. **(02 Marks)**

Consider the following database tables that store data about books, students and books borrowed by the students in the ATI

Books

Book_ID	B_Title	Author	Availability
B001	Workshop Eng.	Saman	Yes
B002	Computer Science	Nuwan	No
B003	Workshop Eng	Kumar	No
B004	Eng Mathematics	Saman	Yes
B004	Networking	Saman	Yes

Barrow

B_Date	Book_ID	Stu_ID
12/07/2018	B002	S004
23/07/2018	B002	S005
23/07/2018	B001	S005
24/07/2018	B004	S001

Students

Stu_ID	Stu_Name	Department
S001	Susantha	Civil
S002	Sonali	Elec
S003	Anne	QS
S004	Shiva	Elec
S005	Kusum	Mech
S006	Sugath	Elec

i. Identify primary key(s) of above three tables **(03 Marks)**

ii. Identify foreign key for the Student and Book Tables **(02 Marks)**

iii. Identify data types for the fields Date, Book_ID and Stu_ID **(03 Marks)**

iv. If Sugath borrowed Computer Science book which table(s) need(s) to be updated.

(02 Marks)

v. Write SQL quires to get following outputs

a. To display bookID barrowed by Kusum **(02 Marks)**

b. To display Book_ID and Stu_ID borrowed between 15/07/1018 and 23/07/2018 **(02 Marks)**

c. To display Book_ID and B_Title written by Saman **(02 Marks)**

d. To display numbers of the borrowed books **(02 Marks)**

Question 06

i. List three (03) advantages and three (03) disadvantages of social Medias. **(06 Marks)**

ii. Define the term "Internet Service Provider (ISP)" by giving two examples. **(03 Marks)**

iii. Name a most suitable Protocol for the following tasks **(04 Marks)**

a. To transform a web page

b. Online banking/shopping or secure transaction

c. To send/receive an e-mail

d. To remote logging

iv. List three (03) precautions that can be taken to prevent viruses/unauthorized accessing or harmful programs which are coming from the internet. **(03 Marks)**

v. What do you mean by a search engine? Give two examples. **(02 Marks)**

vi. What do you mean by a website? Write a programming language that can be used to create a web page **(02 Marks)**